DELIVERABLE NO. 1
THE PROBLEM AND ITS BACKGROUND

Finding a place to stay turns out to be one of the problems that students have to solve, especially those who live far away from their campus. The task of finding a suitable boarding house or dormitory can be complicated since students have to look for offers, double-check the conditions of places, availability, prices, and much more. It is not rare to see students asking for recommendations from friends, searching through various websites, and even visiting multiple places to find the most suitable condition that fits their needs and budget.

To address the issue and remove some stress from the students, the researchers introduced DormFinder, a web application that would help them find available boarding houses or dormitories. The tool will include information about the accommodation and its conditions, such as location, price, amenities, and availability. The property owners can also use DormFinder to post information about their available places.

The main goal of the project is to provide the user with an easy finding of the suitable place to stay. Thus, the researchers wanted to create a convenient tool for students searching for a place and an opportunity for property owners to post their offers on the web application.

1.1 Rationale

Searching for suitable off-campus housing is often a tedious, manual process for students who need a place to live near their area of study. Dorm finder systems are typically recognized as a web-based, location-aware system that catalogues, sort, and filter options for students who wish to simplify the process of finding and securing a place to stay in their region. Dorm finder systems act as a database of specific details for each property, including prices, safety standards, distances, and further services accessible to the student, filtering the best options according to the student's budget and personal preferences while avoiding the need to manually visit several places to look for vacancies.

As previously mentioned, dorm finder systems are web-based resources designed to help students find a suitable place to live near their campus of study. In the literature review of international publications, Leo et al. (2024) describe dorm finder web systems as tools that assist students in searching and selecting suitable housing options based on factors such as cost, location, available facilities, and safety. These systems provide organized information that can help students compare available accommodations according to their needs and preferences..

Meanwhile, local studies have observed the impact of such programs on the current issue of informal boarding in the Philippine context. Leona et al. (2023) offer a conceptual framework for a web-based dorm and reservation system as an application designed to mediate the process of finding off-campus housing while eliminating the need for manual interventions. Local students encounter difficulties in finding a reliable and safe place with competitive rates, and the dorm reservation systems aim to resolve this issue by connecting boarders with less competitive and unlisted properties. The systems would offer occupancy listings, location tracking, and direct contact options for both parties to request and provide services.

Conceptually, the proposed system would be a dormitory and vacancy finder web application designed to offer a convenient and accessible way for students to browse the listings of student accommodations near their campus of enrollment. The system will be utilized as a resource to sort and filter the search for places to stay, offering both occupancy listings and direct messaging for a convenient and straightforward experience for the students and the property owners.

1.2 **Project Context**

This software project entitled "DormFinder: A Web-Based Dormitory Search and Management System" is a web-based system that helps students find dormitories based on their location, budget, and required features. Dormitory and Property owners can also manage their available rooms and property information. The main issue the project is trying to solve is the time-consuming task of finding good places to live. The project will involve students and dormitory or property owners who will share information and provide feedback on the system. Students will have a way to find suitable dormitories, while property owners can display and manage their available rooms for people who want to rent them. The researchers are introducing DormFinder to make the process of looking for and listing accommodations easier and more organized. The project will be built by gathering requirements, designing the system, building it, testing it, and putting it into use. The system will be implemented in the area chosen by the researchers and completed according to the planned schedule.

1.3 **Statement of the Problem**

This capstone project aims to develop DormFinder: A Web-Based Dormitory Search and Management System that helps students find suitable dormitories and available rooms to live in. DormFinder will let users search by location, budget, and other features. Finding a place can take a long time because useful details about boarding houses, apartments, and dormitories are often scattered and not found in a single place. Property owners also need a platform to share and manage information about their dormitories and available rooms.

Specifically, this capstone project seeks to answer the following questions:

- What challenges do students face while searching for dormitories?
- What information do students regard as important when looking for dormitories?
- What features does DormFinder need to help users find places to live?
- How can DormFinder help property owners display and manage details about the rooms and places they have?
- How can the DormFinder system be evaluated for its functionality, ease of use, reliability, and convenience?

1.4 **Objectives of the Study**

This capstone project entitled "DormFinder: A Web-Based Dormitory Search and Management System" aims to develop and implement a web application to aid students, property owners, and school administrators in searching, managing, and securing off-campus student accommodations.

Specifically, this capstone project aims to:

- Gather and analyze system requirements based on the needs and current accommodation search practices of students and property owners.
- Design and develop the DormFinder web application featuring interactive location maps, property filtering, room availability tracking, and direct messaging between students and landlords.
- Integrate a filtering mechanism to help students find suitable dormitories based on rental cost, location, amenities, room availability, and other preferences.
- Evaluate the developed web application through system and usability testing to verify that it meets the required software specifications and user expectations.
- Establish a software maintenance plan to handle system security, database backups, and long-term application updates.

1.5 Significance of the Study

The development of the DormFinder web application will help improve the process of searching, booking, and managing accommodations for student boarders. The application will make it easier and more convenient for students to find available dormitories and boarding houses that fit their needs. It will also help property owners manage and promote their available accommodations online.

The following individuals and entities will benefit from the research and the developed application.

The Students / Boarders. The application will help students find available dormitories and boarding houses without having to search manually. They can view important information such as the location, price, amenities, and other details of the accommodation before making a booking request.

The Dormitory / Property Owners. The application will help dormitory and property owners promote their available rooms and provide information about their accommodations. It will also help them manage booking requests in a more organized way.

The University / School Administration. The application can help the school administration by providing students with a reference for available accommodations outside the school. It can make it easier for students to find information about possible places to stay near the school.

The Researchers. This study will help the researchers improve their knowledge and skills in research and web application development. It will also give them experience in creating a system based on the needs of its users.

Future Researchers. This study can be used as a reference by future researchers who want to develop similar systems related to dormitories, boarding houses, rental services, or student accommodations.

1.6 Scope and Limitations

The current capstone project titled “DormFinder: A Web-Based Dormitory Search and Management System” is concerned with the provision of information about dormitories where students may live. The information concerning dormitories, rooms, prices for rent, room specifications, facilities provided, and room availability is provided by the system. The dormitory or property owner may provide and manage information about his/her property and rooms available for rent. The system enables students to look for suitable dormitories and submit booking/reservation requests for available rooms. The property owner may receive and manage booking/reservation requests sent by users. The system does not have online payment, deposit, or agreement between the property owner and renters. The researchers will not take responsibility for the state of the dormitory, agreements between users and property owners, and any transactions that occur outside the system.

1.7 Definition of Terms

The following terms are defined operationally as used in this study.

Accommodation / Listing. The record of an available dormitory, boarding house, or apartment room stored in the system, which includes details such as monthly rent, location, room layout, amenities, and safety features.

Student / Boarder. A person enrolled in an educational institution who uses the web platform to search for nearby housing, filter options based on budget and preferences, and send inquiries to property owners.

DormFinder Web Application. The software system developed in this study, accessed through a web browser, that enables students to discover housing and allows property owners to manage rental listings.

Direct Messaging. The communication feature within the system that allows students to send inquiry messages directly to property owners regarding room availability, visits, or rental details.

Location Mapping. The interactive map feature embedded in the web application that displays the physical location of an accommodation and its distance relative to the university campus.

Property Owner / Landlord. The individual or business entity authorized to register, post, update, and manage property details, room pricing, and vacancy statuses on the platform.

Real-Time Room Availability. The vacancy status of a room or bed capacity displayed on a property listing, which updates whenever a property owner modifies room records in the system.

Search and Filtering. The feature that allows students to narrow down housing choices based on specific parameters such as price range, distance from school, property type, and included amenities.